PROJECT PRESENTATION

By: K.Chandran

Edge-Based Visual Odometry using Distance Transform Maps Replicating Distance Transform Visual Odometry

AM5011 : Virtual Reality Engineering

DELIVERABLES

- Core Tracking Pipeline — Canny edge detection + distance-transform–based geometric reprojection error

(Detects edges in every incoming frame and measures alignment error via a precomputed distance map instead of pixel intensity, making tracking robust to lighting changes.)

- Weighted Edge Alignment — 5-class edge-pixel weighting with coarse-to-fine pyramid optimization

(Classifies edge pixels by local structure and gradually shifts optimization weight from dominant to minor edge groups across pyramid levels for more accurate pose refinement.)

- Local Mapping — Sliding-window joint optimization of edge depths & camera poses

(Maintains a small window of recent keyframes and jointly refines their depth estimates and poses together, correcting drift that frame-to-frame tracking alone can't fix.)

- Monocular + RGB-D Modes — Random-depth initialization; direct depth-sensor extension

(Supports both a camera-only setup (bootstrapping depth randomly and refining it over time) and a depth-camera setup (using sensor depth directly, skipping initialization))

- Quantitative Evaluation — ATE / RPE benchmarking on TUM RGB-D dataset vs. baseline VO methods

(Measures how closely the estimated camera trajectory matches ground truth, comparing results against established methods like ORB-SLAM2 and DSO)

- Ablation & Efficiency Study — Validate weighted-alignment gains; measure real-time (fps) performance

(Tests the pipeline with and without the weighted alignment scheme to confirm it actually improves accuracy, and benchmarks runtime to check real-time feasibility)

TIMELINE

Foundations — Camera model, SE(3) pose representation, Canny edge detection & distance-transform pipeline

Core Tracking — Geometric reprojection error formulation & Huber-weighted pose optimization

Weighted Edge Alignment — 5-class edge weighting with coarse-to-fine pyramid scheme

Monocular Initialization — Random-depth bootstrap & keyframe selection logic

Local Mapping — Sliding-window joint optimization of depth & poses (LM + marginalization)

RGB-D Extension — Depth-sensor integration & pipeline validation

Evaluation & Reporting — TUM RGB-D benchmarking, ablation study, efficiency analysis, final write-up

THANK YOU

PROJECT REVIEW -1

By: K.Chandran

Replicating and Extending Edge-Based Visual Odometry

AM5011 : Virtual Reality Engineering

7th October 2026

LITERATURE REVIEW

- Robust and efficient edge-based visual odometry (2022):
  - Uses lightning-fast 2D distance maps to track all room outlines purely by geometry, making it immune to lighting changes and able to run at 70 FPS on a single cheap camera.
- EdgeVO (2023):
  - Discards ~80% of useless or noisy outline pixels using an information-scoring algorithm to track only the most geometrically stable edges even faster.
- DynaSLAM (2018):
  - Uses deep-learning AI to detect moving humans and cars, completely erasing them and their corner points so the camera tracks only the static background. (point based VO)
- EEDVO (2016):
  - Fuses raw pixel colors with edge distance maps on depth cameras to prevent slipping along lines, but remains fragile to sudden lighting changes.

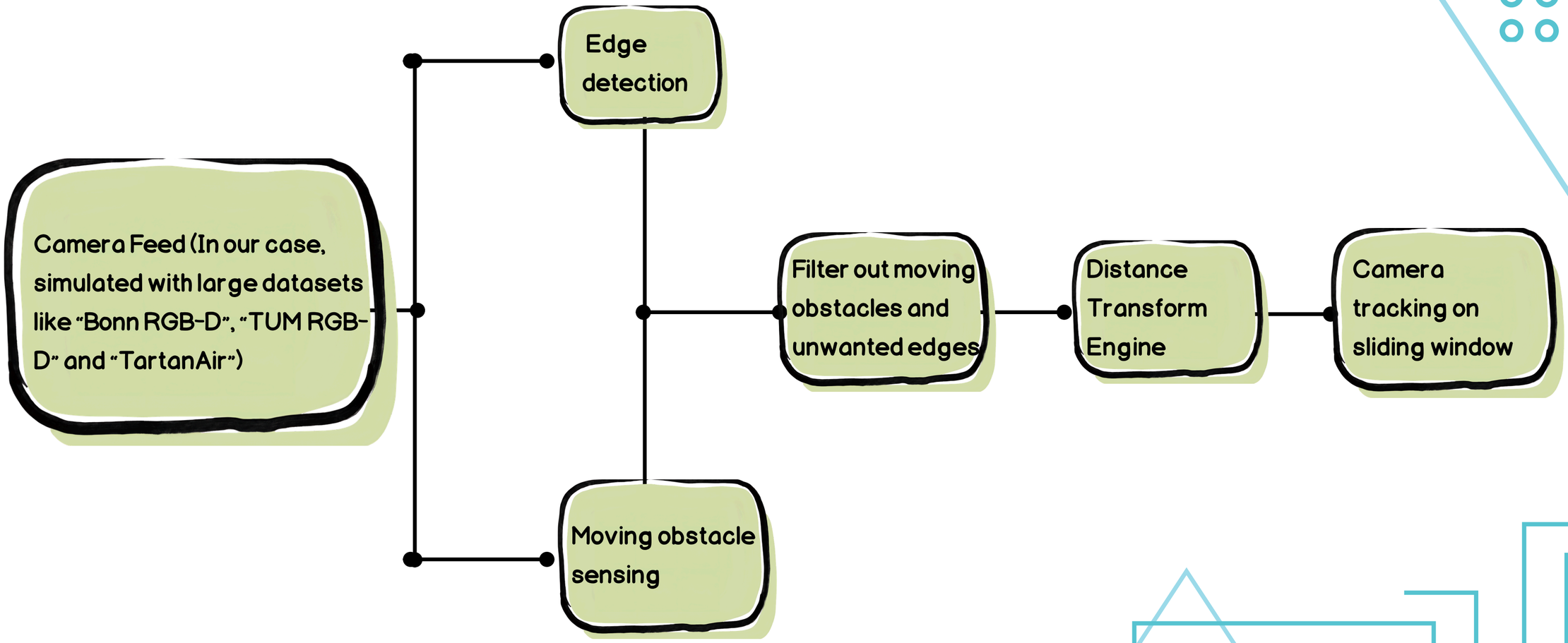
RESEARCH GAP

- DynaSLAM was mentioned before to show a glimpse of the research gaps in edge based VO: there is no method to **filter out moving obstacles** if takes up the majority of the camera feed's screen size.
  - Using light weight AI models to detect moving obstacles and filter out the edges associated with them.

- Moving objects in feed was **never majorly accounted for** in any of the aforementioned papers.
  - EEDVO and EdgeVO clearly states the assumption of a static environment.
  - DTVO (the original paper) explained the use of Hubert loss, which works when very small obstacles are moving in the environment.

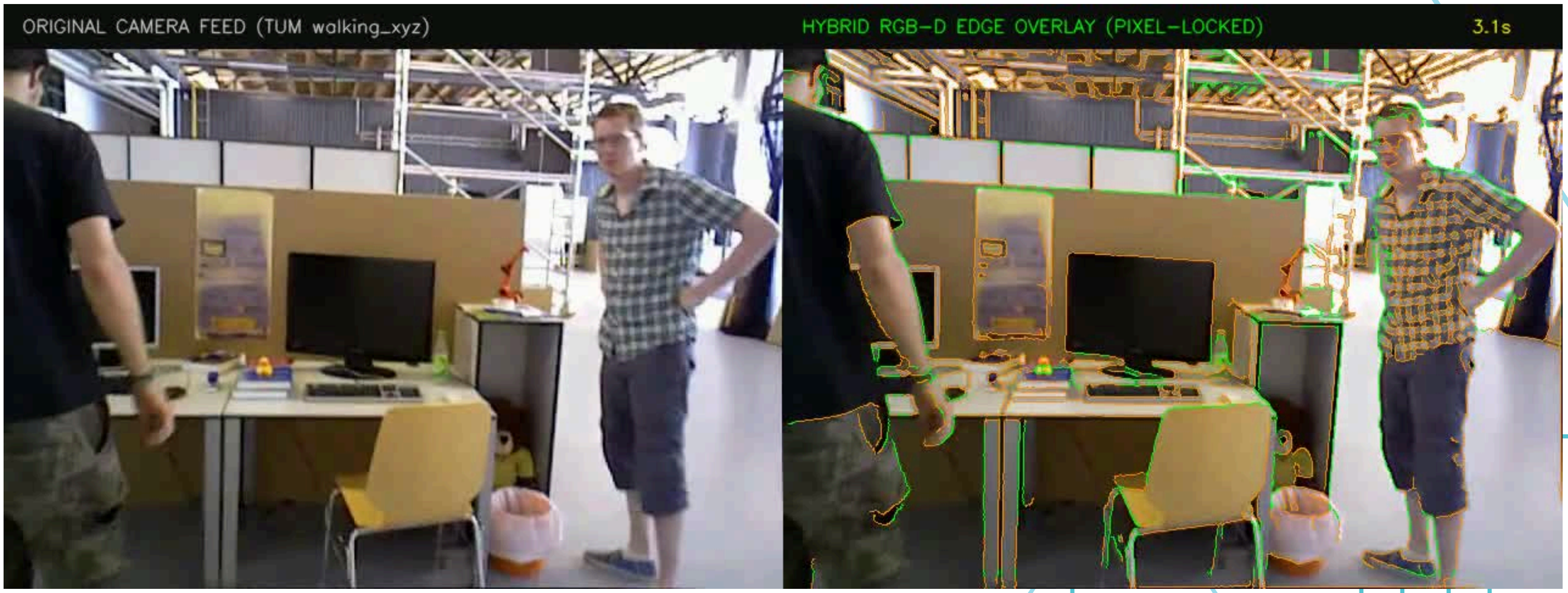
Summary: No proposed VO method to work in real time, on poorly textured environments with moving objects in the camera feed

PROJECT ARCHITECTURE

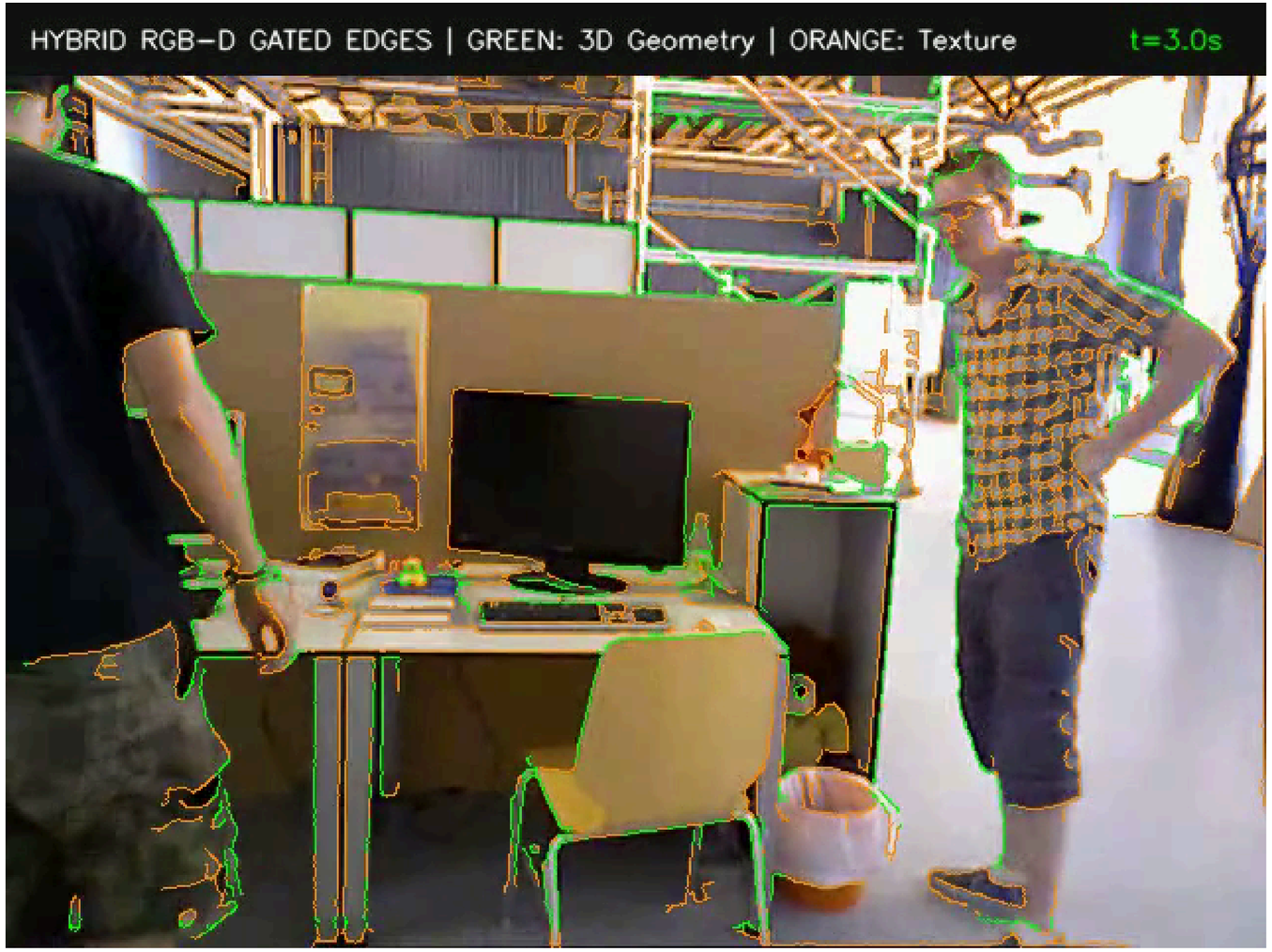


CURRENT PROGRESS

- All required datasets have been downloaded
- Hybrid RGB-D bilateral edge detection has been implemented:
  - Direct implementation of "RGB-D Edge Detection and Edge-Based Registration (IEEE Xplore (DOI: 10.1109/IROS.2013.6696970))
- Chosen method over Canny edge detection as slightly more reliable when implemented over varying shadows and other lighting features



NECESSITY FOR RGB-D



RESOURCE REQUIRED, NEXT STEPS

Resources Required

- Any decently powerful laptop for testing of the edge VO systems
- RGB-D video datasets
- GPU powered computer/server for model training (google collab)

(Fully self sufficient project, doesn't require any lab resources)

Next steps

- Development of a model which detects large moving obstacles in the video feed and filters them out.
- Distance transform map implementation
- Live camera tracking
- Tuning and ablation studies

THANK YOU